# BLESSED C. MADUKOMA

Website |Email: blessedmadukoma@gmail.com |Phone: +2349030609267 | LinkedIn: Madukoma Blessed | GitHub | Blog

## SUMMARY

A product engineer with 3+ years of software engineering, data science and machine learning experience building enterprise scalable systems, predictive models and statistical solutions that improve operational efficiency. Having a strong foundation in software engineering, I specialize in delivering reliable and efficient solutions, applying data analysis and machine learning solutions using Python and cloud services. Demonstrated expertise in transforming complex problems into actionable business insights across international development projects.

## PROFESSIONAL EXPERIENCE

**Carnegie Mellon University Africa** – Kigali, Rwanda

*Graduate Research Assistant, Languages and Spatial Technologies Laboratory* **Sep 2025 – Present**

- Developing machine learning-based anomaly detection framework for athletics performance analysis under Prof. Prasenjit Mitra, implementing statistical and ensemble methods to identify performance patterns indicative of potential doping violations
- Conducting systematic literature review of AI applications in anti-doping, identifying research gaps and unexplored methodologies for next-generation detection systems in track and field athletics

**Ministry of Infrastructure (MININFRA)** – Kigali, Rwanda

*Product Engineer (Contract)* **May 2025 – Sep 2025**

- Architected Rwanda's first infrastructure intelligence platform integrating real estate and transportation data streams through Change Data Capture and Kafka event streaming, with scalable architecture designed for additional ministry data sources.
- Developed Vue.js dashboard with 12+ tailored trend analysis for each infrastructure sector (transportation, and real estate), consolidating cross-ministry data to attract foreign investment.
- Contributed to technical decision-making across 3 development phases, documenting architectural choices and design rationale through documentation pieces and authoring 2+ technical blogs.

**Huza HR** – Kigali, Rwanda

*Software Engineer* **Jun 2025 – Sep 2025**

- Performed code review for 2 team members across the ATS recruitment platform, reviewing 5+ pull requests and establishing coding standards for job seeker and recruiter workflow systems.
- Built an email notification system handling 2+ email templates for the recruitment workflow, handling candidate communications, recruiter alerts, and internal notifications with 99% delivery tracking.
- Built responsive pages, core recruitment interface in NextJS and backend services in NodeJS for the ATS system, establishing foundation to onboard 100+ job seekers and 10+ recruiters in Rwanda's market.

**The UPANZI Network/Cy-LAB** – Kigali, Rwanda

*Graduate Research Assistant* **Sep 2024 – Dec 2024**

- Migrated African datasets hosting platform to Tailwind CSS, reducing bundle size by 30%, improving customization, increasing development speed by 25%, and ensuring data privacy standards while maintaining sensitive datasets.
- Optimized platform performance using Redux, reducing load time by 40%, minimizing database queries and improving user experience.
- Collaborated with cross-functional teams using Agile Scrum methodologies (e.g., weekly standups) and Git-based version control systems, fostering a 15% increase in team productivity and ensuring timely project delivery.

**PLATNOVA** - Remote

*Golang Backend Engineer (Contract)* **Mar 2024 - Aug 2024**

- Revamped the transaction receipt layout using Tailwind CSS, improving customization and enhancing clarity of transaction details, which led to a 30% increase in customer satisfaction and a 15% reduction in support inquiries.
- Accelerated operational efficiency, and optimized cash flow management for vault payment settlements by implementing CRON job, resulting in a 20% reduction in payment processing time and increasing transaction processing speed by 40%.
- Executed and launched a lifestyle-stays package, driving customer engagement and increasing revenue by 15%.

**21ST CENTURY TECHNOLOGIES** – Lagos, Nigeria

**Software Engineer** **Dec 2022 - Nov 2023**

- Implemented Continuous Integration and Continuous Deployment (CI/CD) pipelines to improve software deployment by 70%, reducing release times by 40% and achieving faster resolution times by 10%, saving costs by 40%.
- Engineered data and machine learning pipelines, ensuring 99.99% uptime by orchestrating DNS service restoration across Tier III and IV data centers, reducing downtime by swiftly implementing failover protocols, and enhancing system reliability.
- Delivered a detailed report system integrating data visualization dashboards and communication feedback loops between teams and managers, improving management decision-making and customer satisfaction by 45%.

## PUBLICATIONS

- Wafula, S., **Madukoma, B.** (2025). "Unmasking COVID-19 Vulnerability in Nigeria: Mapping Risks Beyond Urban Hotspots." *Women in Machine Learning Workshop @ NeurIPS 2025*. San Diego, CA. PDF | arXiv

## PROJECTS

**Telco Customer Churn Analysis** [GitHub] **Carnegie Mellon University | 2024**

- Produced 15 actionable retention strategies by designing an ML pipeline using Python (Pandas, Scikit-learn, Plotly), applying K-means clustering, Random Forest, SMOTE, and PCA to analyze customer segments, achieving 26.92% churn reduction prediction.

**Economic Growth & Air Transport Development Analysis (World Bank Project)** [GitHub] **Carnegie Mellon University | 2024**

- Constructed an advanced analytical framework using Python libraries such as Pandas and Matplotlib to dissect four decades of World Bank economic datasets; identified key trends impacting the U.S. air transport sector by clarifying 75.8% variance.

**Core Sentiment Analysis** [GitHub] **Core Data Engineers | 2024**

- Developed a stock prediction tool utilizing Docker and Apache Airflow to orchestrate ETL workflows that extract Wikipedia pageviews as sentiment indicators for major tech companies. Implemented PostgreSQL for data storage, Streamlit for visualization, and GitHub Actions for CI/CD automation to ensure reliable testing and deployment.
- **Technologies used:** Python, Docker, Apache Airflow, PostgreSQL, Streamlit, and GitHub Actions.

**ZeptoMail Go Package** [GitHub] **ZeptoMail Library | 2023**

- Pioneered ZeptoMail's Golang integration for transactional email functionality, improving communication for the Go ecosystem.

## EDUCATION

**CARNEGIE MELLON UNIVERSITY** – Kigali, Rwanda
*Master of Science in Information Technology* **Jul 2024 – May 2026**

- **Specialization**: Software Engineering, Big Data and Applied Machine Learning
- **Thesis Research Focus:** Performance Anomaly Detection in Sports - Developing machine learning models to identify anomalous athletic performances using statistical methods and performance pattern analysis. ***(In Progress)***

**BABCOCK UNIVERSITY –** Ogun, Nigeria
*Bachelor of Science in Software Engineering,* First Class **Jul 2018 – Jun 2022**

## SKILLS & CERTIFICATIONS

**Languages & Core:** Python, Go, TypeScript, JavaScript, SQL, Php | FastAPI, Scikit-learn, Pandas, NumPy, Vue/Nuxt, React/NextJS, NodeJS/Express, NestJS

**Cloud & DevOps:** AWS, GCP, Heroku - Deployment | Airflow, Docker, Terraform | Git, GitHub, GitHub Actions & BitBucket – CI/CD

**Data and Event Systems:** SQL – PostgreSQL, MySQL; NoSQL – MongoDB | Kafka, RabbitMQ | Data Modeling & Design

**Data Science & Analysis:** Time Series Analysis, ML Models (Regression, Classification, Clustering), ETL Pipelines, Predictive Analytics, Data Visualization, Feature Engineering, Model Evaluation & Hyperparameter Tuning

**Computer Science:** Algorithms & Data Structures, System Design, Distributed Systems.

**Certifications:** Data Engineering Essentials, Introduction to Data Engineering, Docker Training, Agile with Jira.

## RELEVANT COURSEWORK

**Machine Learning in Production** **Carnegie Mellon University | Fall 2025**

Developed proficiency in production-ready ML system design, covering the lifecycle from prototype models to deployed products. Developed hands-on wisdom in MLOps practices, system architecture for ML-enabled applications, and responsible AI engineering. Applied quality assurance from model testing to system validation, requirements engineering, and deployment strategies emphasizing safety, security, and fairness. Built practical experience integrating ML models into end-user products while building patterns for scalability, reliability, and operational excellence in real-world scenarios.

**Engineering Data-Intensive Scalable Systems** **Carnegie Mellon University | Spring 2025**

Gained expertise in designing cloud-native distributed systems focusing on microservices, RESTful API design, asynchronous messaging patterns, and data consistency models. Developed hands-on experience through multi-stage AWS cloud projects, using containerization (Docker & Kubernetes), Kafka event streaming, circuit breakers, authentication systems, and NoSQL data stores while applying patterns for scalability, reliability, and high availability.

**Data Structures and Algorithms (DSA)** **Carnegie Mellon University | Spring 2025**

Implemented advanced data structures and algorithms (including Red-Black Trees, priority queues, and custom sparse matrices) across multiple C++ projects, optimizing memory usage, achieving O(log n) operations, and enhanced performance for applications. Some projects include a prioritized queue and dynamic stack-based toll booth simulation that modeled real-time vehicle prioritization, processed Poisson-distributed arrivals, and optimized resource allocation without relying on Standard Template Library. A high-precision eye-tracking analytics engine that processed raw gaze data to identify fixation points, generate visual scan paths, and quantify attentional patterns using customized spatial-temporal distinctness algorithms.

## INTERESTS & ACHIEVEMENTS

- Chairman, Academic Innovation Sub-Committee of the Research Committee in Carnegie Mellon University Africa.
- Volunteer, Google DevFest Kigali, December 2024.
- Executive President, Babcock University Computer Club, Babcock University, Nigeria, 2022.